**97** Identify the **correct** answer.

(1) Dipole moment of $NF_3$ is greater than that of $NH_3$.

(2) Three canonical forms can be drawn for $CO_3^{2-}$ ion.

(3) Three resonance structures can be drawn for ozone.

(4) $BF_3$ has non-zero dipole moment.

**98** The pair of lanthanoid ions which are diamagnetic is

(1) $Gd^{3+}$ and $Eu^{3+}$

(2) $Pm^{3+}$ and $Sm^{3+}$

(3) $Ce^{4+}$ and $Yb^{2+}$

(4) $Ce^{3+}$ and $Eu^{2+}$

**99** Given below are two statements :

**Statement I :** $\left[Co(NH_3)_6\right]^{3+}$ is a homoleptic complex whereas $\left[Co(NH_3)_4Cl_2\right]^{+}$ is a heteroleptic complex.

**Statement II :** Complex $\left[Co(NH_3)_6\right]^{3+}$ has only one kind of ligands but $\left[Co(NH_3)_4Cl_2\right]^{+}$ has more than one kind of ligands.

In the light of the above statements, choose the *correct* answer from the options given below:

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

**100** Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is:

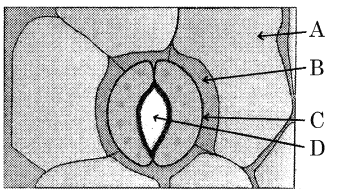
(Given : Molar mass of Cu : 63 g $mol^{-1}$, 1F = 96487 C)

(1) 31.5 g (2) 0.0315 g

(3) 3.15 g (4) 0.315 g

## Botany : Section-A (Q. No. 101 to 135)

**101** In the given figure, which component has thin outer walls and highly thickened inner walls?




(1) A (2) B

(3) C (4) D

**102** List of endangered species was released by-

(1) FOAM (2) IUCN

(3) GEAC (4) WWF

**103** The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called;

(1) Semi-conservative method

(2) Sustainable development

(3) *in-situ* conservation

(4) Biodiversity conservation

**104** Which one of the following can be explained on the basis of Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive.

B. Alleles do not show any expression and both the characters appear as such in $F_2$ generation.

C. Factors occur in pairs in normal diploid plants.

D. The discrete unit controlling a particular character is called factor.

E. The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

(1) B, C and D only

(2) A, B, C, D and E

(3) A, B and C only

(4) A, C, D and E only

**105** The lactose present in the growth medium of bacteria is transported to the cell by the action of:
(1) Permease
(2) Polymerase
(3) Beta-galactosidase
(4) Acetylase

**106** Match List I with List II

| | **List I** | | **List II** |
|---|---|---|---|
| A. | *Clostridium butylicum* | I. | Ethanol |
| B. | *Saccharomyces cerevisiae* | II. | Streptokinase |
| C. | *Trichoderma polysporum* | III. | Butyric acid |
| D. | *Streptococcus* sp. | IV. | Cyclosporin-A |

Choose the correct answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-I, C-III, D-II
(3) A-III, B-I, C-II, D-IV
(4) A-II, B-IV, C-III, D-I

**107** The equation of Verhulst-Pearl logistic growth is

$$\frac{dN}{dt} = rN\left[\frac{K-N}{K}\right].$$

From this equation, $K$ indicates:
(1) Carrying capacity
(2) Population density
(3) Intrinsic rate of natural increase
(4) Biotic potential

**108** A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end;
(1) Inducer, Repressor, Structural gene
(2) Promotor, Structural gene, Terminator
(3) Repressor, Operator gene, Structural gene
(4) Structural gene, Transposons, Operator gene

**109** Formation of interfascicular cambium from fully developed parenchyma cells is an example for
(1) Dedifferentiation
(2) Maturation
(3) Differentiation
(4) Redifferentiation

**110** Match List I with List II

| | **List I** | | **List II** |
|---|---|---|---|
| A. | *Rhizopus* | I. | Mushroom |
| B. | *Ustilago* | II. | Smut fungus |
| C. | *Puccinia* | III. | Bread mould |
| D. | *Agaricus* | IV. | Rust fungus |

Choose the correct answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-I, B-III, C-II, D-IV

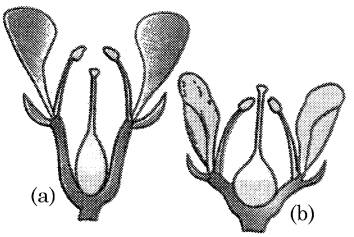
**111** What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism?
A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.
B. It may get integrated into the genome of the recipient.
C. It may multiply and be inherited along with the host DNA.
D. The alien piece of DNA is not an integral part of chromosome.
E. It shows ability to replicate.
Choose the correct answer from the options given below:
(1) B and C only
(2) A and E only
(3) A and B only
(4) D and E only

**112** Match List I with List II

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Nucleolus | I. | Site of formation of glycolipid |
| B. | Centriole | II. | Organization like the cartwheel |
| C. | Leucoplasts | III. | Site for active ribosomal RNA synthesis |
| D. | Golgi apparatus | IV. | For storing nutrients |

Choose the correct answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-III, C-I, D-IV

**113** Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figures (a) and (b)




(1) (a) Perigynous; (b) Epigynous
(2) (a) Perigynous; (b) Perigynous
(3) (a) Epigynous; (b) Hypogynous
(4) (a) Hypogynous; (b) Epigynous

**114** Spindle fibers attach to kinetochores of chromosomes during
(1) Anaphase (2) Telophase
(3) Prophase (4) Metaphase

**115** These are regarded as major causes of biodiversity loss:
A. Over exploitation
B. Co-extinction
C. Mutation
D. Habitat loss and fragmentation
E. Migration
Choose the correct option:
(1) A, B and E only
(2) A, B and D only
(3) A, C and D only
(4) A, B, C and D only

**116** Lecithin, a small molecular weight organic compound found in living tissues, is an example of:
(1) Glycerides
(2) Carbohydrates
(3) Amino acids
(4) Phospholipids

**117** Identify the set of correct statements:
A. The flowers of *Vallisneria* are colourful and produce nectar.
B. The flowers of waterlily are not pollinated by water.
C. In most of water-pollinated species, the pollen grains are protected from wetting.
D. Pollen grains of some hydrophytes are long and ribbon like.
E. In some hydrophytes, the pollen grains are carried passively inside water.
Choose the correct answer from the options given below:
(1) A, C, D and E only
(2) B, C, D and E only
(3) C, D and E only
(4) A, B, C and D only

**118** The cofactor of the enzyme carboxypeptidase is:
(1) Flavin (2) Haem
(3) Zinc (4) Niacin

**119** Inhibition of Succinic dehydrogenase enzyme by malonate is a classical example of:
(1) Competitive inhibition
(2) Enzyme activation
(3) Cofactor inhibition
(4) Feedback inhibition

**120** Which of the following is an example of actinomorphic flower?
(1) *Pisum* (2) *Sesbania*
(3) *Datura* (4) *Cassia*

**121** Given below are two statements:
**Statement I :** Chromosomes become gradually visible under light microscope during leptotene stage.
**Statement II :** The begining of diplotene stage is recognized by dissolution of synaptonemal complex.
In the light of the above statements, choose the correct answer from the options given below:
(1) Statement I is true but Statement II is false
(2) Statement I is false but Statement II is true
(3) Both Statement I and Statement II are true
(4) Both Statement I and Statement II are false

**122** A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny?

(1) Only pink flowered plants

(2) Red, Pink as well as white flowered plants

(3) Only red flowered plants

(4) Red flowered as well as pink flowered plants

**123** Given below are two statements:

**Statement I :** Parenchyma is living but collenchyma is dead tissue.

**Statement II :** Gymnosperms lack xylem vessels but presence of xylem vessels is the characteristic of angiosperms.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

**124** Given below are two statements:

**Statement I :** Bt toxins are insect group specific and coded by a gene *cry* IAc.

**Statement II :** Bt toxin exists as inactive protoxin in *B. thuringiensis*. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

**125** Which one of the following is not a criterion for classification of fungi?

(1) Mode of spore formation

(2) Fruiting body

(3) Morphology of mycelium

(4) Mode of nutrition

**126** The capacity to generate a whole plant from any cell of the plant is called:

(1) Differentiation

(2) Somatic hybridization

(3) Totipotency

(4) Micropropagation

**127** Which of the following are required for the dark reaction of photosynthesis?

A. Light

B. Chlorophyll

C. $CO_2$

D. ATP

E. NADPH

Choose the correct answer from the options given below:

(1) C, D and E only

(2) D and E only

(3) A, B and C only

(4) B, C and D only

**128** Tropical regions show greatest level of species richness because

A. Tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.

B. Tropical environments are more seasonal.

C. More solar energy is available in tropics.

D. Constant environments promote niche specialization.

E. Tropical environments are constant and predictable.

Choose the correct answer from the options given below:

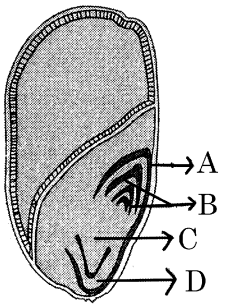
(1) A, B and E only

(2) A, B and D only

(3) A, C, D and E only

(4) A and B only

**129** Identify the part of the seed from the given figure which is destined to form root when the seed germinates.




(1) C (2) D

(3) A (4) B

**130** How many molecules of ATP and NADPH are required for every molecule of $CO_2$ fixed in the Calvin cycle?

(1) 3 molecules of ATP and 3 molecules of NADPH

(2) 3 molecules of ATP and 2 molecules of NADPH

(3) 2 molecules of ATP and 3 molecules of NADPH

(4) 2 molecules of ATP and 2 molecules of NADPH

**131** In a plant, black seed color (BB/Bb) is dominant over white seed color (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?

(1) Bb (2) BB/Bb

(3) BB (4) bb

**132** Hind II always cuts DNA molecules at a particular point called recognition sequence and it consists of:

(1) 4 bp (2) 10 bp

(3) 8 bp (4) 6 bp

**133** Bulliform cells are responsible for

(1) Increased photosynthesis in monocots.

(2) Providing large spaces for storage of sugars.

(3) Inward curling of leaves in monocots.

(4) Protecting the plant from salt stress.

**134** Match List I with List II

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Two or more alternative forms of a gene | I. | Back cross |
| B. | Cross of $F_1$ progeny with homozygous recessive parent | II. | Ploidy |
| C. | Cross of $F_1$ progeny with any of the parents | III. | Allele |
| D. | Number of chromosome sets in plant | IV. | Test cross |

Choose the correct answer from the options given below:

(1) A-III, B-IV, C-I, D-II

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-I, C-III, D-IV

**135** Auxin is used by gardeners to prepare weed-free lawns. But no damage is caused to grass as auxin

(1) does not affect mature monocotyledonous plants.

(2) can help in cell division in grasses, to produce growth.

(3) promotes apical dominance.

(4) promotes abscission of mature leaves only.

**Botany : Section-B (Q. No. 136 to 150)**

**136** Match List I with List II

| | List I | | List II |
|---|---|---|---|
| A. | Rose | I. | Twisted aestivation |
| B. | Pea | II. | Perigynous flower |
| C. | Cotton | III. | Drupe |
| D. | Mango | IV. | Marginal placentation |

Choose the correct answer from the options given below:

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-III, C-IV, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-I, B-II, C-III, D-IV

**137** The DNA present in chloroplast is:

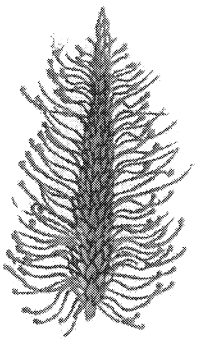
(1) Linear, single stranded

(2) Circular, single stranded

(3) Linear, double stranded

(4) Circular, double stranded

**138** Identify the correct description about the given figure:



(1) Cleistogamous flowers showing autogamy.

(2) Compact inflorescence showing complete autogamy.

(3) Wind pollinated plant inflorescence showing flowers with well exposed stamens.

(4) Water pollinated flowers showing stamens with mucilaginous covering.

**139** Match List I with List II

| | List I | | List II |
|---|---|---|---|
| A. | Robert May | I. | Species-Area relationship |
| B. | Alexander von Humboldt | II. | Long term ecosystem experiment using out door plots |
| C. | Paul Ehrlich | III. | Global species diversity at about 7 million |
| D. | David Tilman | IV. | Rivet popper hypothesis |

Choose the correct answer from the options given below:

(1) A-I, B-III, C-II, D-IV

(2) A-III, B-IV, C-II, D-I

(3) A-II, B-III, C-I, D-IV

(4) A-III, B-I, C-IV, D-II

**140** Given below are two statements:

**Statement I :** In $C_3$ plants, some $O_2$ binds to RuBisCO, hence $CO_2$ fixation is decreased.

**Statement II :** In $C_4$ plants, mesophyll cells show very little photorespiration while bundle sheath cells do not show photorespiration.

In the light of the above statements, choose the *correct* answer from the options given below:

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

**141** Identify the step in tricarboxylic acid cycle, which does not involve oxidation of substrate.

(1) Succinyl-CoA $\rightarrow$ Succinic acid

(2) Isocitrate $\rightarrow$ $\alpha$-ketoglutaric acid

(3) Malic acid $\rightarrow$ Oxaloacetic acid

(4) Succinic acid $\rightarrow$ Malic acid

**142** Match List I with List II

| | List I | | List II |
|---|---|---|---|
| A. | Citric acid cycle | I. | Cytoplasm |
| B. | Glycolysis | II. | Mitochondrial matrix |
| C. | Electron transport system | III. | Intermembrane space of mitochondria |
| D. | Proton gradient | IV. | Inner mitochondrial membrane |

Choose the correct answer from the options given below:

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-II, D-I
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-I, C-IV, D-III

**143** Which of the following are fused in somatic hybridization involving two varieties of plants?

(1) Protoplasts
(2) Pollens
(3) Callus
(4) Somatic embryos

**144** Match List I with List II

| | List I | | List II |
|---|---|---|---|
| A. | GLUT-4 | I. | Hormone |
| B. | Insulin | II. | Enzyme |
| C. | Trypsin | III. | Intercellular ground substance |
| D. | Collagen | IV. | Enables glucose transport into cells |

Choose the correct answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-III, B-IV, C-I, D-II
(3) A-IV, B-I, C-II, D-III
(4) A-I, B-II, C-III, D-IV

**145** Match List I with List II

| | List I (Types of Stamens) | | List II (Example) |
|---|---|---|---|
| A. | Monoadelphous | I. | Citrus |
| B. | Diadelphous | II. | Pea |
| C. | Polyadelphous | III. | Lily |
| D. | Epiphyllous | IV. | China-rose |

Choose the correct answer from the options given below:

(1) A-I, B-II, C-IV, D-III
(2) A-III, B-I, C-IV, D-II
(3) A-IV, B-II, C-I, D-III
(4) A-IV, B-I, C-II, D-III

**146** Read the following statements and choose the set of correct statements:

In the members of Phaeophyceae,

A. Asexual reproduction occurs usually by biflagellate zoospores.
B. Sexual reproduction is by oogamous method only.
C. Stored food is in the form of carbohydrates which is either mannitol or laminarin.
D. The major pigments found are chlorophyll a, c and carotenoids and xanthophyll.
E. Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.

Choose the correct answer from the options given below:

(1) A, C, D and E only
(2) A, B, C and E only
(3) A, B, C and D only
(4) B, C, D and E only

**147** Match List I with List II

| | List I | | List II |
|---|---|---|---|
| A. | Frederick Griffith | I. | Genetic code |
| B. | Francois Jacob & Jacque Monod | II. | Semi-conservative mode of DNA replication |
| C. | Har Gobind Khorana | III. | Transformation |
| D. | Meselson & Stahl | IV. | *Lac* operon |

Choose the correct answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-III, B-IV, C-I, D-II

**Zoology : Section-A (Q. No. 151 to 185)**

**148** Which of the following statement is correct regarding the process of replication in *E.coli*?

(1) The DNA dependent DNA polymerase catalyses polymerization in 5' → 3' as well as 3' → 5' direction.

(2) The DNA dependent DNA polymerase catalyses polymerization in 5' → 3' direction.

(3) The DNA dependent DNA polymerase catalyses polymerization in one direction that is 3' → 5'.

(4) The DNA dependent RNA polymerase catalyses polymerization in one direction, that is 5' → 3'.

**149** Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus, increasing the yield?

(1) Cytokinin

(2) Abscisic acid

(3) Auxin

(4) Gibberellin

**150** In an ecosystem if the Net Primary Productivity (NPP) of first trophic level is

$100x\,(kcal\,m^{-2})\,yr^{-1}$, what would be the GPP (Gross Primary Productivity) of the third trophic level of the same ecosystem?

(1) $10x\,(kcal\,m^{-2})\,yr^{-1}$

(2) $\frac{100x}{3x}\,(kcal\,m^{-2})\,yr^{-1}$

(3) $\frac{x}{10}\,(kcal\,m^{-2})\,yr^{-1}$

(4) $x\,(kcal\,m^{-2})\,yr^{-1}$

**151** Match List I with List II :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Down's syndrome | I. | $11^{th}$ chromosome |
| B. | α-Thalassemia | II. | 'X' chromosome |
| C. | β-Thalassemia | III. | $21^{st}$ chromosome |
| D. | Klinefelter's syndrome | IV. | $16^{th}$ chromosome |

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-I, D-II

(2) A-IV, B-I, C-II, D-III

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-IV, D-I

**152** Match List I with List II :

| | **List I (Sub Phases of Prophase I)** | | **List II (Specific characters)** |
|---|---|---|---|
| A. | Diakinesis | I. | Synaptonemal complex formation |
| B. | Pachytene | II. | Completion of terminalisation of chiasmata |
| C. | Zygotene | III. | Chromosomes look like thin threads |
| D. | Leptotene | IV. | Appearance of recombination nodules |

Choose the correct answer from the options given below :

(1) A-II, B-IV, C-I, D-III

(2) A-IV, B-III, C-II, D-I

(3) A-IV, B-II, C-III, D-I

(4) A-I, B-II, C-IV, D-III

**153** Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?

(1) Low $pCO_2$ and High $H^+$ concentration

(2) Low $pCO_2$ and High temperature

(3) High $pO_2$ and High $pCO_2$

(4) High $pO_2$ and Lesser $H^+$ concentration

**154** Match List I with List II :

| List I | | List II |
|---|---|---|
| A. Typhoid | I. | Fungus |
| B. Leishmaniasis | II. | Nematode |
| C. Ringworm | III. | Protozoa |
| D. Filariasis | IV. | Bacteria |

Choose the correct answer from the options given below :

(1) A-III, B-I, C-IV, D-II
(2) A-II, B-IV, C-III, D-I
(3) A-I, B-III, C-II, D-IV
(4) A-IV, B-III, C-I, D-II

**155** Match List I with List II :

| List I | | List II |
|---|---|---|
| A. Expiratory capacity | I. | Expiratory reserve volume + Tidal volume + Inspiratory reserve volume |
| B. Functional residual capacity | II. | Tidal volume + Expiratory reserve volume |
| C. Vital capacity | III. | Tidal volume + Inspiratory reserve volume |
| D. Inspiratory capacity | IV. | Expiratory reserve volume + Residual volume |

Choose the correct answer from the options given below :

(1) A-II, B-I, C-IV, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-II, C-IV, D-I

**156** Which of the following are Autoimmune disorders?

A. Myasthenia gravis
B. Rheumatoid arthritis
C. Gout
D. Muscular dystrophy
E. Systemic Lupus Erythematosus (SLE)

Choose the most appropriate answer from the options given below :

(1) B, C & E only
(2) C, D & E only
(3) A, B & D only
(4) A, B & E only

**157** Given below are two statements :

**Statement I :** In the nephron, the descending limb of loop of Henle is impermeable to water and permeable to electrolytes.

**Statement II :** The proximal convoluted tubule is lined by simple columnar brush border epithelium and increases the surface area for reabsorption.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false
(2) Statement I is false but Statement II is true
(3) Both Statement I and Statement II are true
(4) Both Statement I and Statement II are false

**158** Which of the following is not a steroid hormone?

(1) Progesterone
(2) Glucagon
(3) Cortisol
(4) Testosterone

**159** Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

**Assertion A :** FSH acts upon ovarian follicles in female and Leydig cells in male.

**Reason R :** Growing ovarian follicles secrete estrogen in female while interstitial cells secrete androgen in male human being.

In the light of the above statements, choose the correct answer from the options given below :

(1) A is true but R is false
(2) A is false but R is true
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

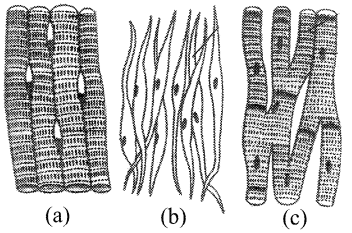
**160** Given below are some stages of human evolution. Arrange them in correct sequence. (Past to Recent)

A. *Homo habilis*
B. *Homo sapiens*
C. *Homo neanderthalensis*
D. *Homo erectus*

Choose the correct sequence of human evolution from the options given below :

(1) C-B-D-A (2) A-D-C-B
(3) D-A-C-B (4) B-A-D-C

**161** Three types of muscles are given as a, b and c. Identify the correct matching pair along with their location in human body :




**Name of muscle/location**

(1) (a) Skeletal - Biceps
(b) Involuntary - Intestine
(c) Smooth - Heart.

(2) (a) Involuntary - Nose tip
(b) Skeletal - Bone
(c) Cardiac - Heart.

(3) (a) Smooth - Toes
(b) Skeletal - Legs
(c) Cardiac - Heart.

(4) (a) Skeletal - Triceps
(b) Smooth - Stomach
(c) Cardiac - Heart.

**162** Match List I with List II :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Cocaine | I. | Effective sedative in surgery |
| B. | Heroin | II. | *Cannabis sativa* |
| C. | Morphine | III. | *Erythroxylum* |
| D. | Marijuana | IV. | *Papaver somniferum* |

Choose the correct answer from the options given below :

(1) A-II, B-I, C-III, D-IV
(2) A-III, B-IV, C-I, D-II
(3) A-IV, B-III, C-I, D-II
(4) A-I, B-III, C-II, D-IV

**163** The "Ti plasmid" of *Agrobacterium tumefaciens* stands for

(1) Tumor inducing plasmid
(2) Temperature independent plasmid
(3) Tumour inhibiting plasmid
(4) Tumor independent plasmid

**164** Which one is the correct product of DNA dependent RNA polymerase to the given template?

3'TACATGGCAAATATCCATTCA5'

(1) 5'AUGUACCGUUUAUAGGGAAGU3'
(2) 5'ATGTACCGTTTATAGGTAAGT3'
(3) 5'AUGUACCGUUUAUAGGUAAGU3'
(4) 5'AUGUAAAGUUUAUAGGUAAGU3'

**165** Match List I with List II :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | *Pterophyllum* | I. | Hag fish |
| B. | *Myxine* | II. | Saw fish |
| C. | *Pristis* | III. | Angel fish |
| D. | *Exocoetus* | IV. | Flying fish |

Choose the correct answer from the options given below :

(1) A-IV, B-I, C-II, D-III
(2) A-III, B-II, C-I, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-II, D-IV

**166** Match List I with List II :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Fibrous joints | I. | Adjacent vertebrae, limited movement |
| B. | Cartilaginous joints | II. | Humerus and Pectoral girdle, rotational movement |
| C. | Hinge joints | III. | Skull, don't allow any movement |
| D. | Ball and socket joints | IV. | Knee, help in locomotion |

Choose the correct answer from the options given below :

(1) A-II, B-III, C-I, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-IV, B-II, C-III, D-I
(4) A-I, B-III, C-II, D-IV

**167** The flippers of the Penguins and Dolphins are the example of the

(1) Convergent evolution

(2) Divergent evolution

(3) Adaptive radiation

(4) Natural selection

**168** Match List I with List II :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | α-1 antitrypsin | I. | Cotton bollworm |
| B. | Cry IAb | II. | ADA deficiency |
| C. | Cry IAc | III. | Emphysema |
| D. | Enzyme replacement therapy | IV. | Corn borer |

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-I, D-II

(2) A-II, B-IV, C-I, D-III

(3) A-II, B-I, C-IV, D-III

(4) A-III, B-I, C-II, D-IV

**169** Match List I with List II :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Axoneme | I. | Centriole |
| B. | Cartwheel pattern | II. | Cilia and flagella |
| C. | Crista | III. | Chromosome |
| D. | Satellite | IV. | Mitochondria |

Choose the correct answer from the options given below :

(1) A-II, B-IV, C-I, D-III

(2) A-II, B-I, C-IV, D-III

(3) A-IV, B-III, C-II, D-I

(4) A-IV, B-II, C-III, D-I

**170** Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R :

**Assertion A :** Breast-feeding during initial period of infant growth is recommended by doctors for bringing a healthy baby.

**Reason R :** Colostrum contains several antibodies absolutely essential to develop resistance for the new born baby.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) A is correct but R is not correct.

(2) A is not correct but R is correct.

(3) Both A and R are correct and R is the correct explanation of A.

(4) Both A and R are correct but R is NOT the correct explanation of A.

**171** Which of the following is not a component of Fallopian tube?

(1) Infundibulum

(2) Ampulla

(3) Uterine fundus

(4) Isthmus

**172** Which of the following statements is incorrect?

(1) Bio-reactors are used to produce small scale bacterial cultures.

(2) Bio-reactors have an agitator system, an oxygen delivery system and foam control system.

(3) A bio-reactor provides optimal growth conditions for achieving the desired product.

(4) Most commonly used bio-reactors are of stirring type.

173 Match List I with List II :

| | List I | | List II |
|---|---|---|---|
| A. | Pons | I. | Provides additional space for Neurons, regulates posture and balance. |
| B. | Hypothalamus | II. | Controls respiration and gastric secretions. |
| C. | Medulla | III. | Connects different regions of the brain. |
| D. | Cerebellum | IV. | Neuro secretory cells |

Choose the correct answer from the options given below :

(1) A-I, B-III, C-II, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-IV, C-II, D-I

174 Match List I with List II :

| | List I | | List II |
|---|---|---|---|
| A. | Lipase | I. | Peptide bond |
| B. | Nuclease | II. | Ester bond |
| C. | Protease | III. | Glycosidic bond |
| D. | Amylase | IV. | Phosphodiester bond |

Choose the correct answer from the options given below :

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-I, C-III, D-II
(3) A-IV, B-II, C-III, D-I
(4) A-III, B-II, C-I, D-IV

175 Which of the following is not a natural/traditional contraceptive method?

(1) Lactational amenorrhea
(2) Vaults
(3) Coitus interruptus
(4) Periodic abstinence

176 Following are the stages of pathway for conduction of an action potential through the heart:

A. AV bundle
B. Purkinje fibres
C. AV node
D. Bundle branches
E. SA node

Choose the correct sequence of pathway from the options given below :

(1) B-D-E-C-A (2) E-A-D-B-C
(3) E-C-A-D-B (4) A-E-C-B-D

177 Match List I with List II :

| | List I | | List II |
|---|---|---|---|
| A. | Pleurobrachia | I. | Mollusca |
| B. | Radula | II. | Ctenophora |
| C. | Stomochord | III. | Osteichthyes |
| D. | Air bladder | IV. | Hemichordata |

Choose the correct answer from the options given below :

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-III, C-II, D-I
(3) A-IV, B-II, C-III, D-I
(4) A-II, B-I, C-IV, D-III

178 Match List I with List II :

| | List I | | List II |
|---|---|---|---|
| A. | Common cold | I. | *Plasmodium* |
| B. | Haemozoin | II. | Typhoid |
| C. | Widal test | III. | Rhinoviruses |
| D. | Allergy | IV. | Dust mites |

Choose the correct answer from the options given below :

(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-II, B-IV, C-III, D-I
(4) A-I, B-III, C-II, D-IV

**179** Consider the following statements :

A. Annelids are true coelomates

B. Poriferans are pseudocoelomates

C. Aschelminthes are acoelomates

D. Platyhelminthes are pseudocoelomates

Choose the correct answer from the options given below :

(1) C only (2) D only

(3) B only (4) A only

**180** In both sexes of cockroach, a pair of jointed filamentous structures called anal cerci are present on :

(1) $8^{th}$ and $9^{th}$ segment

(2) $11^{th}$ segment

(3) $5^{th}$ segment

(4) $10^{th}$ segment

**181** Given below are two statements :

**Statement I :** The presence or absence of hymen is not a reliable indicator of virginity.

**Statement II :** The hymen is torn during the first coitus only.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

**182** Which one of the following factors will not affect the Hardy-Weinberg equilibrium?

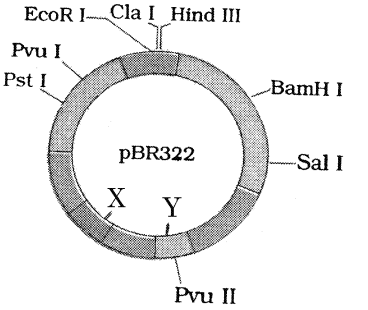
(1) Gene migration

(2) Constant gene pool

(3) Genetic recombination

(4) Genetic drift

**183** The following diagram showing restriction sites in *E.coli* cloning vector pBR322. Find the role of '*X*' and '*Y*' genes :




(1) The gene 'X' is for protein involved in replication of Plasmid and '*Y*' for resistance to antibiotics.

(2) Gene '*X*' is responsible for recognition sites and '*Y*' is responsible for antibiotic resistance.

(3) The gene '*X*' is responsible for resistance to antibiotics and '*Y*' for protein involved in the replication of Plasmid.

(4) The gene '*X*' is responsible for controlling the copy number of the linked DNA and '*Y*' for protein involved in the replication of Plasmid.

**184** Match List I with List II :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | Non-medicated IUD | I. | Multiload 375 |
| B. | Copper releasing IUD | II. | Progestogens |
| C. | Hormone releasing IUD | III. | Lippes loop |
| D. | Implants | IV. | LNG-20 |

Choose the correct answer from the options given below :

(1) A-IV, B-I, C-II, D-III

(2) A-III, B-I, C-IV, D-II

(3) A-III, B-I, C-II, D-IV

(4) A-I, B-III, C-IV, D-II

**185** Following are the stages of cell division :

A. Gap 2 phase

B. Cytokinesis

C. Synthesis phase

D. Karyokinesis

E. Gap 1 phase

Choose the correct sequence of stages from the options given below :

(1) B-D-E-A-C (2) E-C-A-D-B

(3) C-E-D-A-B (4) E-B-D-A-C

## Zoology : Section-B (Q. No. 186 to 200)

**186** Choose the correct statement given below regarding juxta medullary nephron.

(1) Loop of Henle of juxta medullary nephron runs deep into medulla.

(2) Juxta medullary nephrons outnumber the cortical nephrons.

(3) Juxta medullary nephrons are located in the columns of Bertini.

(4) Renal corpuscle of juxta medullary nephron lies in the outer portion of the renal medulla.

**187** The following are the statements about non-chordates :

A. Pharynx is perforated by gill slits.

B. Notochord is absent.

C. Central nervous system is dorsal.

D. Heart is dorsal if present.

E. Post anal tail is absent.

Choose the most appropriate answer from the options given below :

(1) B, D & E only

(2) B, C & D only

(3) A & C only

(4) A, B & D only

**188** Given below are two statements :

**Statement I :** Mitochondria and chloroplasts are both double membrane bound organelles.

**Statement II :** Inner membrane of mitochondria is relatively less permeable, as compared to chloroplast.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

**189** Regarding catalytic cycle of an enzyme action, select the correct sequential steps :

A. Substrate enzyme complex formation.

B. Free enzyme ready to bind with another substrate.

C. Release of products.

D. Chemical bonds of the substrate broken.

E. Substrate binding to active site.

Choose the correct answer from the options given below :

(1) B, A, C, D, E

(2) E, D, C, B, A

(3) E, A, D, C, B

(4) A, E, B, D, C

**190** Match List I with List II :

| | **List I** | | **List II** |
|---|---|---|---|
| A. | P wave | I. | Heart muscles are electrically silent. |
| B. | QRS complex | II. | Depolarisation of ventricles. |
| C. | T wave | III. | Depolarisation of atria. |
| D. | T-P gap | IV. | Repolarisation of ventricles. |

Choose the correct answer from the options given below :

(1) A-II, B-III, C-I, D-IV

(2) A-IV, B-II, C-I, D-III

(3) A-I, B-III, C-IV, D-II

(4) A-III, B-II, C-IV, D-I

**191** Given below are two statements :

**Statement I :** Bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced.

**Statement II :** Both bone marrow and thymus provide micro environments for the development and maturation of T-lymphocytes.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

**192** Match List I with List II :

| | List I | | List II |
|---|---|---|---|
| A. | Unicellular glandular epithelium | I. | Salivary glands |
| B. | Compound epithelium | II. | Pancreas |
| C. | Multicellular glandular epithelium | III. | Goblet cells of alimentary canal |
| D. | Endocrine glandular epithelium | IV. | Moist surface of buccal cavity |

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-I, D-II
(2) A-II, B-I, C-IV, D-III
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-III, C-I, D-II

**193** Given below are two statements :

**Statement I :** The cerebral hemispheres are connected by nerve tract known as corpus callosum.

**Statement II :** The brain stem consists of the medulla oblongata, pons and cerebrum.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

**194** Match List I with List II :

| | List I | | List II |
|---|---|---|---|
| A. | RNA polymerase III | I. | snRNPs |
| B. | Termination of transcription | II. | Promotor |
| C. | Splicing of Exons | III. | Rho factor |
| D. | TATA box | IV. | SnRNAs, tRNA |

Choose the correct answer from the options given below :

(1) A-III, B-IV, C-I, D-II
(2) A-IV, B-III, C-I, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-II, C-IV, D-I

**195** Match List I with List II :

| | List I | | List II |
|---|---|---|---|
| A. | Mesozoic Era | I. | Lower invertebrates |
| B. | Proterozoic Era | II. | Fish & Amphibia |
| C. | Cenozoic Era | III. | Birds & Reptiles |
| D. | Paleozoic Era | IV. | Mammals |

Choose the correct answer from the options given below :

(1) A-I, B-II, C-IV, D-III
(2) A-III, B-I, C-IV, D-II
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-I, C-II, D-IV

**196** Given below are two statements :

**Statement I :** Gause's competitive exclusion principle states that two closely related species competing for different resources cannot exist indefinitely.

**Statement II :** According to Gause's principle, during competition, the inferior will be eliminated. This may be true if resources are limiting.

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is true but Statement II is false.
(2) Statement I is false but Statement II is true.
(3) Both Statement I and Statement II are true.
(4) Both Statement I and Statement II are false.

**197** Match List I with List II :

| | List I | | List II |
|---|---|---|---|
| A. | Exophthalmic goiter | I. | Excess secretion of cortisol, moon face & hyperglycemia |
| B. | Acromegaly | II. | Hypo-secretion of thyroid hormone and stunted growth. |
| C. | Cushing's syndrome | III. | Hyper secretion of thyroid hormone & protruding eye balls. |
| D. | Cretinism | IV. | Excessive secretion of growth hormone. |

Choose the correct answer from the options given below :

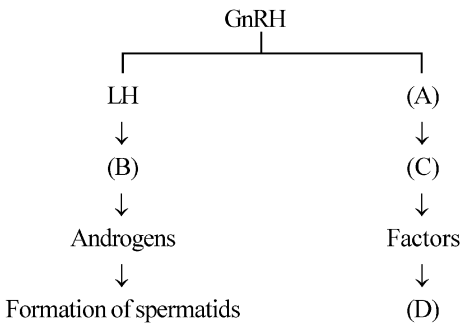
(1) A-III, B-IV, C-II, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-I, B-III, C-II, D-IV

(4) A-IV, B-II, C-I, D-III

**198** Identify the correct option (A), (B), (C), (D) with respect to spermatogenesis.




(1) FSH, Sertoli cells, Leydig cells, spermatogenesis.

(2) ICSH, Leydig cells, Sertoli cells, spermatogenesis.

(3) FSH, Leydig cells, Sertoli cells, spermiogenesis

(4) ICSH, Interstitial cells, Leydig cells, spermiogenesis.

**199** As per ABO blood grouping system, the blood group of father is $B^+$, mother is $A^+$ and child is $O^+$. Their respective genotype can be

A. $I^Bi$ / $I^Ai$ / ii

B. $I^BI^B$ / $I^AI^A$ / ii

C. $I^AI^B$ / $iI^A$ / $I^Bi$

D. $I^Ai$ / $I^Bi$ / $I^Ai$

E. $iI^B$ / $iI^A$ / $I^AI^B$

Choose the most appropriate answer from the options given below :

(1) C & B only (2) D & E only

(3) A only (4) B only

**200** Match List I with List II related to digestive system of cockroach.

| | List I | | List II |
|---|---|---|---|
| A. | The structures used for storing of food. | I. | Gizzard |
| B. | Ring of 6-8 blind tubules at junction of foregut and midgut. | II. | Gastric Caeca |
| C. | Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut. | III. | Malpighian tubules |
| D. | The structures used for grinding the food. | IV. | Crop |

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-III, B-II, C-IV, D-I

(3) A-IV, B-II, C-III, D-I

(4) A-I, B-II, C-III, D-IV